

LIPHENG PRUM

Lecturer of Robotics and Automation

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RESEARCH INTERESTS

My research interests are robot learning, dexterous in-hand manipulation, vision-guided robot control, computer vision, robotics and intelligent systems.

TECHNICAL SKILLS

Robot Learning: ISAAC LAB/SIM

AI & Machine Learning: PyTorch, reinforcement learning dexterous hand & locomotion, deep learning, computer vision

Robotics & Automation: PLC programming (Keyence, Mitsubishi), industrial robotics (6-axis, SCARA), HMI design, STM32CubeIDE, Embedded C

IoT & Embedded Systems: LoRaWAN, MQTT, ESP32, STM32, Raspberry Pi, Node-RED, Arduino

EDUCATION

MS – Robotics and Computational Intelligence Systems **2020 – 2023**

King Mongkut's Institute of Technology Ladkrabang (KMITL), Bangkok, Thailand

GPA: 3.88 / 4.0 (36 credits) Thesis Oral Defense: Grade G (Good)

Thesis: *Finding Picking Point Using Straight Line Detection of Label Attachment Machine*

Relevant coursework: Machine Learning, Artificial Intelligence, Big Data Analytics, Internet of Things, Modern Control Theory, PID Controller Theory/Design/Tuning, Applications of Nonlinear Control to Robotics, Introduction to Industrial Robotics, and Research Methodology.

BS – Production Engineering **2014 – 2018**

Chaiyaphum Rajabhat University, Thailand

GPA: 3.69 / 4.0 · First Honor Class

CONFERENCE

[1] L. Prum, et al., "Finding Picking Point Using Straight Line Detection of Label Attachment Machine," in *Proc. 26th International Computer Science and Engineering Conference (ICSEC)*, pp. 252–256, 2022. IEEE. DOI: 10.1109/ICSEC56337.2022.

PROFESSIONAL EXPERIENCE

Lecturer of Robotics and Automation **2026 – Present**

Cambodia University of Technology and Science (CamTech), Phnom Penh, Cambodia

- Teaching undergraduate courses in robotics and automation. Developing curriculum incorporating hands-on project-based learning and research methodology.

Senior IoT Engineer **2024 – 2026**

CMED Construction Company, Phnom Penh, Cambodia

- Led multiple proof-of-concept R&D projects from concept to working prototype, spanning IoT sensor networks, computer vision, and access control systems. Technologies: Python, Raspberry Pi, MQTT, STM32, Node-RED, Django REST Framework, PyTorch, LangGraph.

Senior Production Engineer (NPI) **2023 – 2024**

SVI Cambodia Company, Phnom Penh, Cambodia

- Led production engineering for new product introduction in SMT/box-build electronics manufacturing. Created manufacturing documentation, optimized production lines, and drove process improvements.

Automation Engineer

2018 – 2023

Cal-Comp Electronics (Thailand) Public Co., Ltd., Samutsakorn, Thailand

- Designed and deployed PLC-based control systems with industrial robots (6-axis, SCARA) integrated via RS232/RS485/TCP-IP. Developed firmware for ESP32/ATMega328 microcontrollers and HMI interfaces.
- This role directly motivated my research direction: recurring failures of rule-based vision systems under changing factory conditions led me to pursue learning-based perception for robust robotic manipulation.

Internships (Production & Maintenance Engineering)

2017 – 2018

Denso Cambodia (Production Engineer) & HGST/Western Digital (Asst. Maintenance Engineer), Thailand

- Supported new line transfers, jig/fixture setup, production/quality documentation (SOPs, Control Plans, PFMEA), and process improvement in high-volume electronics manufacturing.

LANGUAGES

Khmer (Native) · Thai (Fluent) · English (proficiency – Study & Research)

HONORS AND AWARDS

- Distinguished Engineering Student Award (2017) – The Engineering Institute of Thailand under His Majesty the King’s Patronage.
- 6th ASEAN Synchrotron Science Camp (2017) – Selected participant.
- Outstanding Achievement in Project Presentation (2017) – Cooperative & Work Integrated Education Program, HGST (Western Digital).

CONTINUING EDUCATION

- Machine Learning and Deep Learning Specialization – Coursera / Deep Learning. AI. Completed foundational and applied deep learning courses covering neural networks, CNNs, sequence models, and optimization techniques.
- Arm Cortex-M Architecture and Software Development Specialization – Coursera (Issued Aug 2025). Covered Arm Cortex-M processor architecture, embedded software development, and low-level system design.
- Hands-on Internet of Things – Coursera / University of Illinois Urbana-Champaign (Issued Mar 2024, Credential ID: 85VB5HJNTFMQ). Practical IoT system design including device connectivity, sensor integration, and data pipelines.

REFERENCES

Prof. John Morris, PhD (University of Sydney)

Former Faculty, School of Engineering, KMITL; currently Faculty of Engineering, Mahasarakham University, Thailand

Relationship: Co-advisor and co-author of published IEEE conference paper

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Dean of Faculty of Engineering

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